Bayesian Statistics (47-747) — Homework #1

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60 points total. Provide R code for all relevant problems.

Problem 1 (10 points): Beta–Binomial with Arcsine Prior

Coin tossed N times with S heads. Parameter of interest: $\theta = P(\text{heads})$.
Likelihood: Binomial. Prior is the arcsine density

$$f(\theta) = \frac{1}{\pi\sqrt{\theta(1-\theta)}}, \qquad 0 < \theta < 1,$$

which is equivalent to $\text{Beta}(\frac{1}{2}, \frac{1}{2})$.

(a) Posterior, mean, and variance

Write your derivation here.

Given that the probability density function of Beta distribution is given by:

$$f(\theta) = \frac{1}{\beta(\alpha_0, \beta_0)} \theta^{\alpha_0 - 1}(1-\theta)^{\beta_0 - 1}, \qquad 0 < \theta < 1,$$

where $\beta(\alpha_0, \beta_0)$ is the Beta function.

The Beta function is defined as:

$$\beta(\alpha_0, \beta_0) = \int_0^1 \theta^{\alpha_0 - 1}(1-\theta)^{\beta_0 - 1} d\theta$$

The prior is given by:

$$\theta \sim \text{Beta}(\alpha_0, \beta_0)$$

with $\alpha_0 = \beta_0 = \frac{1}{2}$.

The likelihood is given by:

$$S \mid \theta \sim \text{Binomial}(N, \theta)$$

Prior: $\theta \sim \text{Beta}(\alpha_0, \beta_0)$ with $\alpha_0 = \beta_0 = \frac{1}{2}$.
Likelihood: $S \mid \theta \sim \text{Binomial}(N, \theta)$.

Posterior:

$$\theta \mid S, N \sim \text{Beta}(...)$$

Posterior mean:

$$\mathbb{E}[\theta \mid S, N] = ...$$

Posterior variance:

$$\mathrm{Var}(\theta \mid S, N) = ...$$

```
# Optional helper functions (edit / extend as needed)
beta_mean <- function(a, b) a / (a + b)
beta_var  <- function(a, b) (a * b) / ((a + b)^2 * (a + b + 1))
```

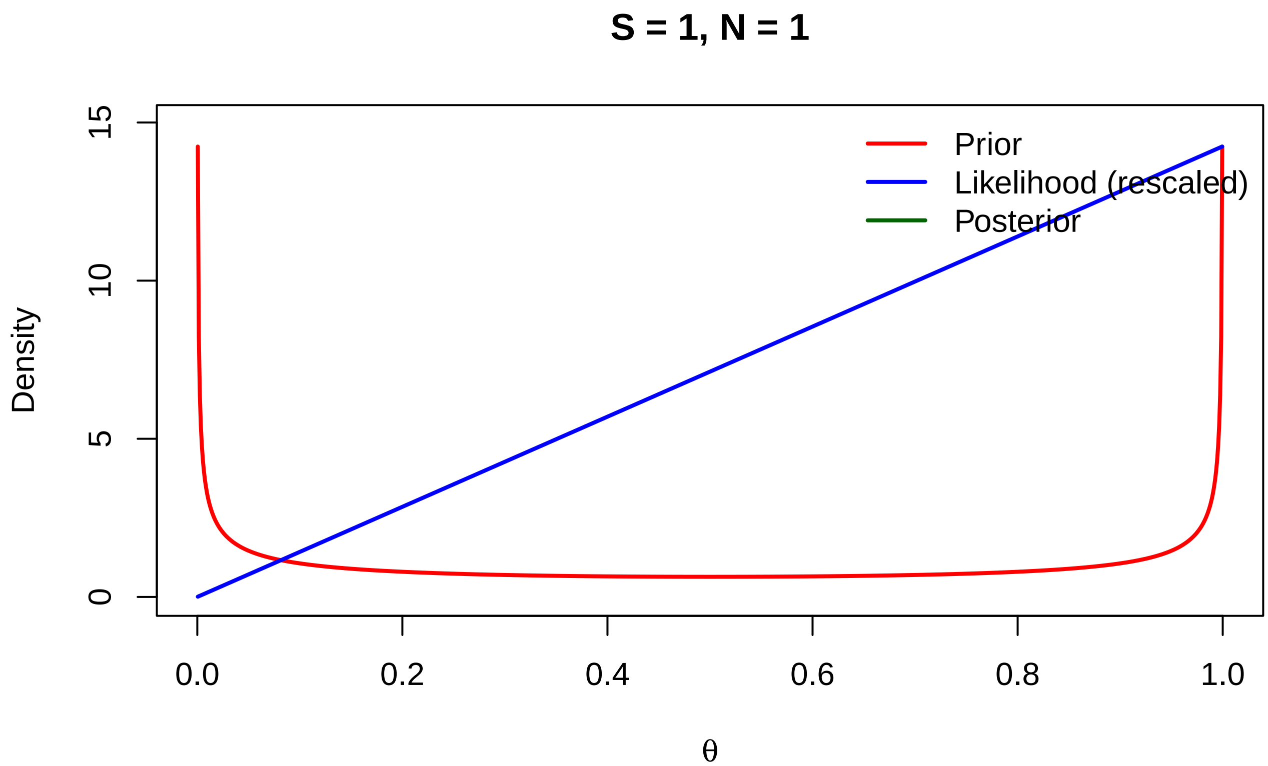
(b) Plot prior, likelihood, and posterior

Overlay prior, likelihood, and posterior for:

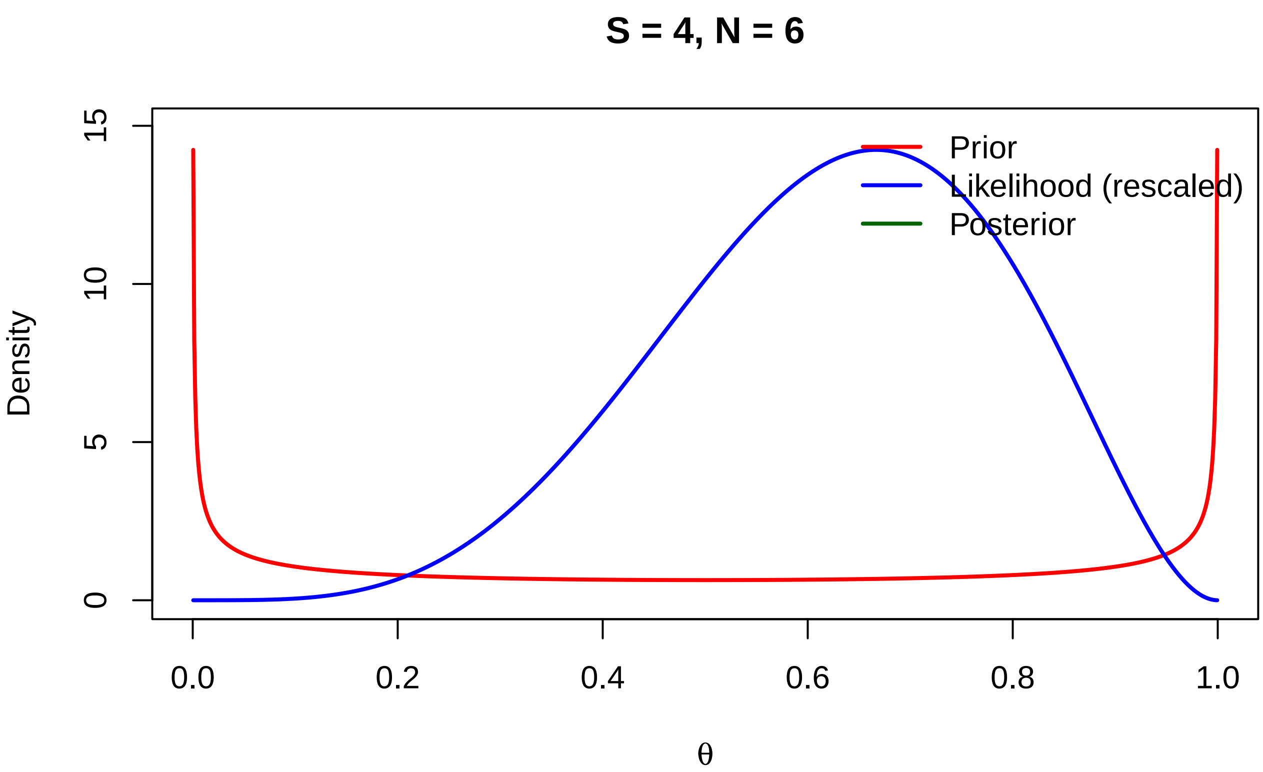
1. $S = 1$, $N = 1$
2. $S = 4$, $N = 6$

```
# TODO: fill in alpha0, beta0 and posterior parameters
plot_beta_binomial <- function(S, N,
                               alpha0 = 0.5, beta0 = 0.5,
                               main = NULL) {
  theta <- ppoints(1000)

  # Prior density
  dens_prior <- dbeta(theta, alpha0, beta0)

  # Likelihood as a function of theta (up to a constant is fine for shape;
  # here we plot the binomial pmf in theta)
  dens_like <- dbinom(S, size = N, prob = theta)

  # TODO: posterior Beta parameters
  alpha_n <- NA   # <- replace
  beta_n  <- NA   # <- replace
  dens_post <- dbeta(theta, alpha_n, beta_n)

  ylim_max <- max(dens_prior, dens_like / max(dens_like) * max(dens_prior, dens_post, na.rm = TRUE),
                  dens_post, na.rm = TRUE) * 1.05

  if (is.null(main)) {
    main <- sprintf("S = %d, N = %d", S, N)
  }

  plot(theta, dens_prior, type = "l", lwd = 2, col = "red",
       ylim = c(0, ylim_max), xlab = expression(theta), ylab = "Density",
       main = main)
  # Rescale likelihood for visual overlay (comment if you prefer raw scale)
  lines(theta, dens_like / max(dens_like) * max(dens_prior, dens_post, na.rm = TRUE),
        lwd = 2, col = "blue")
  lines(theta, dens_post, lwd = 2, col = "darkgreen")
  legend("topright",
         legend = c("Prior", "Likelihood (rescaled)", "Posterior"),
         col = c("red", "blue", "darkgreen"), lwd = 2, bty = "n")
}
```

```r
# Case i: S = 1, N = 1
plot_beta_binomial(S = 1, N = 1)
```




```r
# Case ii: S = 4, N = 6
plot_beta_binomial(S = 4, N = 6)
```



(c) Sequential update with new data (S_1, N_1)

Write your derivation here.

Using the posterior from (a) as the new prior, after observing S_1 heads in N_1 trials:

$$\theta \mid S, N, S_1, N_1 \sim \text{Beta}(...)$$

Problem 2 (10 points): Combining Normal Forecasts for GDP

Let θ be next year's US GDP growth.

(a) Your prior

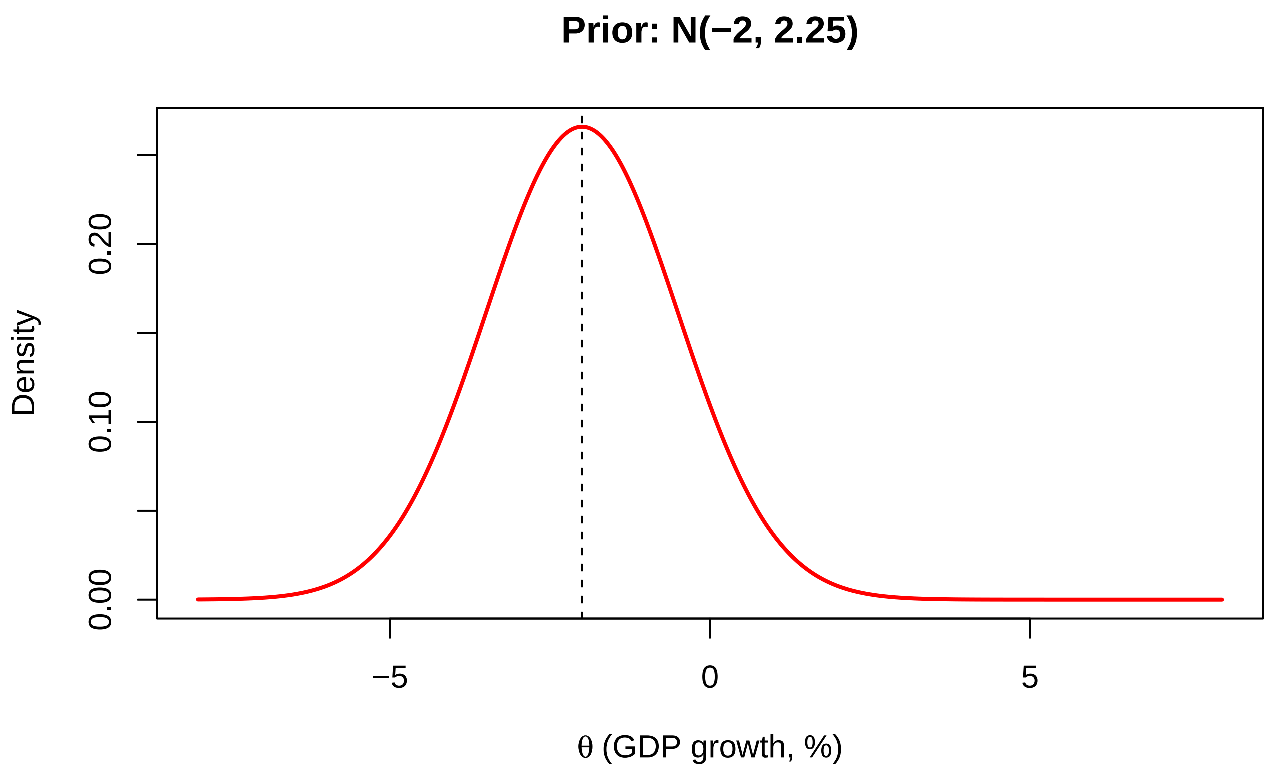
Mean -2% growth, variance 2.25%.
(Work in percentage points: $\mu_0 = -2$, $\sigma_0^2 = 2.25$, so precision $\tau_0 = 1/\sigma_0^2$.)

Write the density with mean μ_0 and precision τ_0:

$$p(\theta) = ...$$

```r
mu0 <- -2
var0 <- 2.25
tau0 <- 1 / var0

theta_grid <- seq(-8, 8, length.out = 1000)
dens0 <- dnorm(theta_grid, mean = mu0, sd = sqrt(var0))
```

```r
plot(theta_grid, dens0, type = "l", lwd = 2, col = "red",
     xlab = expression(theta ~ "(GDP growth, %)"), ylab = "Density",
     main = "Prior: N(-2, 2.25)")
abline(v = mu0, lty = 2)
```



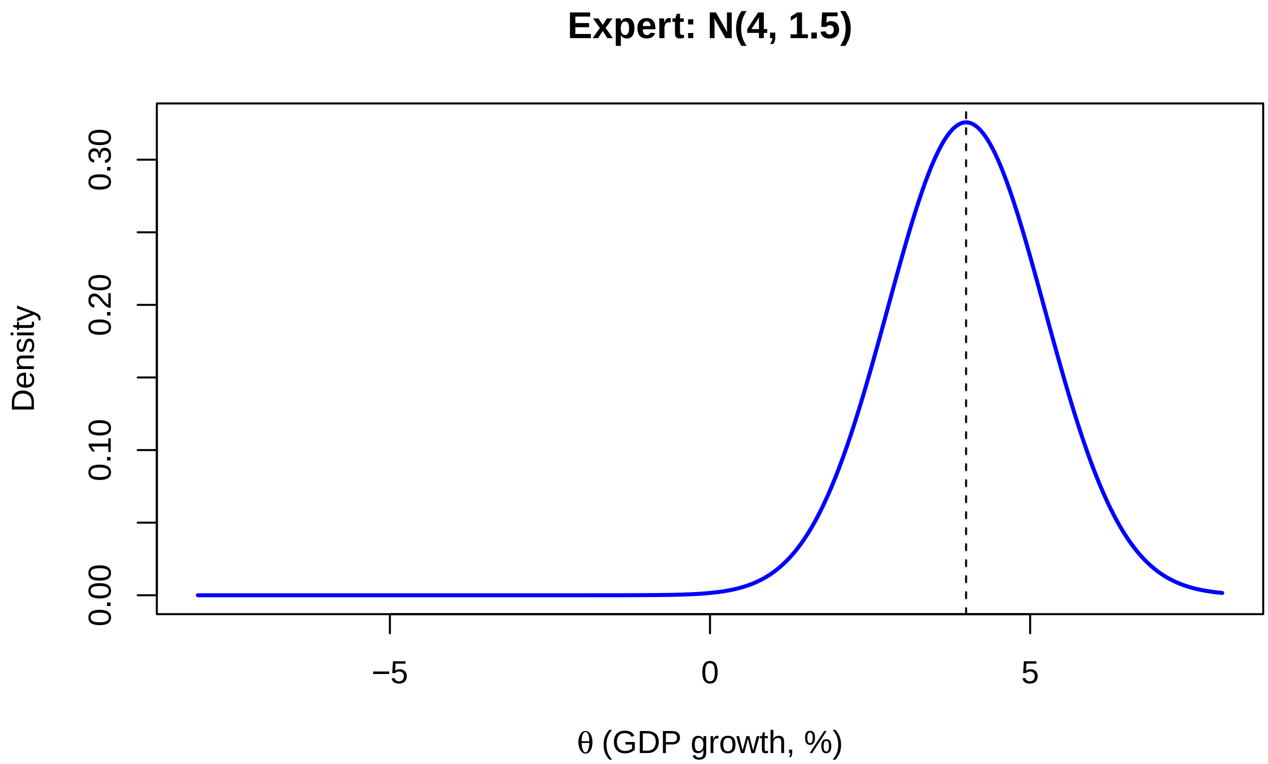

```r
# TODO: print mu0, tau0, and the PDF expression in your write-up
c(mu0 = mu0, tau0 = tau0, var0 = var0)
```

```
##        mu0       tau0       var0
## -2.0000000  0.4444444  2.2500000
```

(b) Expert's information

Mean 4% growth, variance 1.5%. Independent of yours.
Write the density with mean μ and precision τ, and plot it.

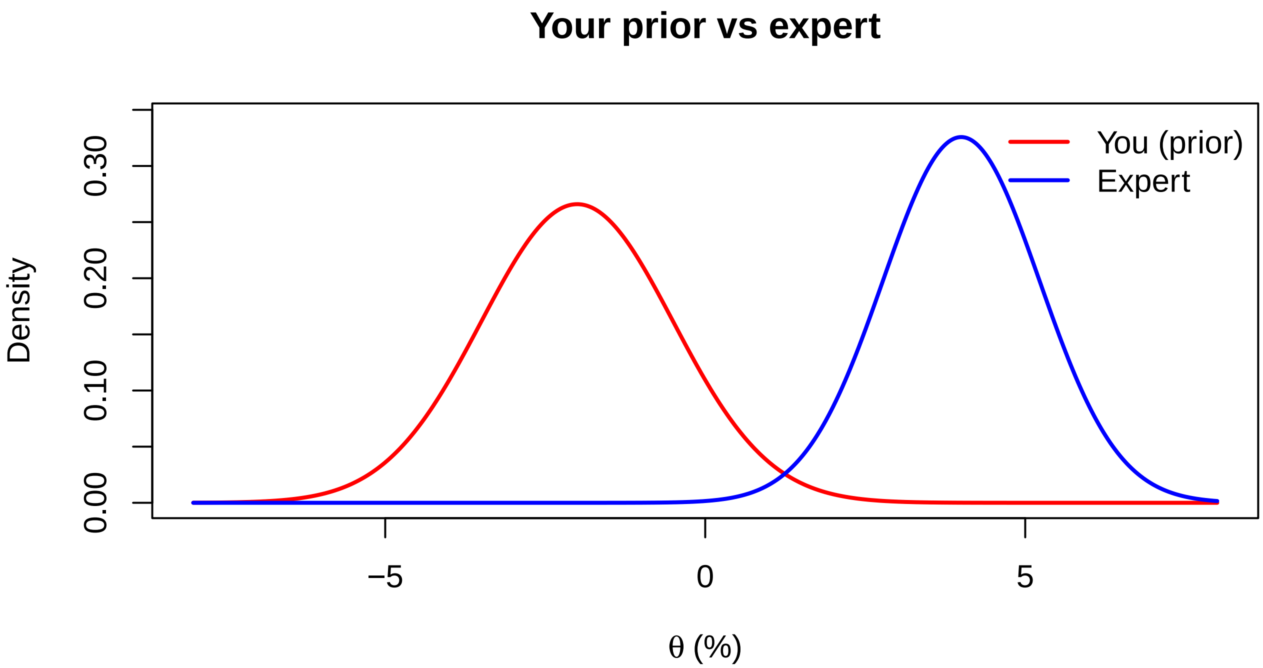
```r
mu  <- 4
var <- 1.5
tau <- 1 / var

dens_exp <- dnorm(theta_grid, mean = mu, sd = sqrt(var))

plot(theta_grid, dens_exp, type = "l", lwd = 2, col = "blue",
     xlab = expression(theta ~ "(GDP growth, %)"), ylab = "Density",
     main = "Expert: N(4, 1.5)")
abline(v = mu, lty = 2)
```




```r
c(mu = mu, tau = tau, var = var)
```

```
##        mu       tau       var
## 4.0000000 0.6666667 1.5000000
```

```r
# Optional: overlay both forecasts
plot(theta_grid, dens0, type = "l", lwd = 2, col = "red",
     ylim = c(0, max(dens0, dens_exp) * 1.05),
     xlab = expression(theta ~ "(%)"), ylab = "Density",
     main = "Your prior vs expert")
lines(theta_grid, dens_exp, lwd = 2, col = "blue")
legend("topright", c("You (prior)", "Expert"),
       col = c("red", "blue"), lwd = 2, bty = "n")
```



(c) Combining the two forecasts

Write your answer in words here.

Hints you may use (optional computation — not required by the prompt):

- Treat the expert as independent Normal "data" / information and update your Normal prior.
- Or form the product of the two Normal densities (precision-weighted average of means).

```
# Optional: precision-weighted combination (for your own check)
# tau_post <- tau0 + tau
# mu_post  <- (tau0 * mu0 + tau * mu) / tau_post
```

Problem 3 (10 points): Bayesian Linear Regression

Model:

$$y_i = \mathbf{x}_i'\beta + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2),$$

with $\beta = (\beta_0, \beta_1)'$ (2×1).

Data:

$$\mathbf{y} = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 8 \\ 10 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

For each prior below, report:

- posterior for β and σ^2

- posterior means and 95% credible intervals
- density plot overlaying **prior**, **likelihood**, and **posterior** for β_1

Data setup

```
y <- c(1, 4, 3, 8, 10)
X <- cbind(1, 0:4)
n <- length(y)
k <- ncol(X)

# OLS quantities (useful for likelihood / diffuse posterior)
XtX <- crossprod(X)
Xty <- crossprod(X, y)
beta_ols <- as.numeric(solve(XtX, Xty))
resid <- as.numeric(y - X %*% beta_ols)
SSE <- sum(resid^2)
s2_ols <- SSE / (n - k)

list(beta_ols = beta_ols, SSE = SSE, s2_ols = s2_ols)
```

```
## $beta_ols
## [1] 0.8 2.2
##
## $SSE
## [1] 6.4
##
## $s2_ols
## [1] 2.133333
```

```
# Skeleton for Normal-Inverse-Gamma / conjugate Normal regression updates.
# Fill in formulas from lecture notes as needed.
#
# Prior: beta | sigma2 ~ N(mu0, sigma2 * Lambda0^{-1})
#        sigma2 ~ IG(nu0/2, nu0*sigma0_sq/2)   [check lecture parameterization]
#
# Returns a list with posterior hyperparameters and summaries for beta1.

summarize_beta1 <- function(mu_n, scale_cov, nu_n, label = "") {
  # Marginal posterior for beta is multivariate-t (after integrating sigma2).
  # TODO: use the correct scale for the marginal Student-t of beta1.
  #
  # Placeholder: Normal approximation using posterior mean of sigma2
  # Replace with exact t-based credible intervals from lecture.
  sigma2_hat <- NA
  mean_b1 <- mu_n[2]
  # se_b1 <- ...
  # ci <- mean_b1 + qt(c(0.025, 0.975), df = ...) * se_b1
  list(
    label = label,
    mean_beta = mu_n,
    mean_beta1 = mean_b1,
    ci95_beta1 = c(NA, NA),
    note = "TODO: replace Normal approx with exact posterior CI from notes"
  )
```

```r
}

plot_beta1_overlay <- function(prior_dens, like_dens, post_dens, grid,
                               main = expression(beta[1])) {
  ymax <- max(prior_dens, like_dens, post_dens, na.rm = TRUE) * 1.05
  plot(grid, prior_dens, type = "l", lwd = 2, col = "red",
       ylim = c(0, ymax), xlab = expression(beta[1]), ylab = "Density",
       main = main)
  lines(grid, like_dens, lwd = 2, col = "blue")
  lines(grid, post_dens, lwd = 2, col = "darkgreen")
  legend("topright", c("Prior", "Likelihood", "Posterior"),
         col = c("red", "blue", "darkgreen"), lwd = 2, bty = "n")
}
```

(a) Diffuse prior

```r
# TODO: implement diffuse / noninformative prior posterior
# Classic reference posterior under flat prior on beta and log(sigma):
#   beta | sigma2, y ~ N(beta_ols, sigma2 * (X'X)^{-1})
#   sigma2 | y ~ IG( ... )  [fill from notes]

mu_n_a <- beta_ols
# Lambda_n_a <- ...
# nu_n_a <- ...
# sigma0_n_a <- ...

# Posterior means / 95% CIs:
# ...

# Plot for beta1:
# b1_grid <- seq(..., ..., length.out = 500)
# plot_beta1_overlay(...)
```

Answers (a):

- Posterior of β, σ^2: ...
- Means / 95% CIs: ...

(b) Informative prior

$\mu_0 = (1, 2)'$, $\Lambda_0 = I_2$, $\nu_0 = 5$, $\sigma_0^2 = 0.6$.

```r
mu0 <- c(1, 2)
Lambda0 <- diag(2)
nu0 <- 5
sigma0_sq <- 0.6

# TODO: conjugate update
# Lambda_n <- Lambda0 + X'X
# mu_n <- solve(Lambda_n, Lambda0 %*% mu0 + X'y)
# nu_n <- nu0 + n
# update scale for sigma2 using quadratic forms from notes
```

Answers (b): ...

(c)–(e) g-priors

Same hyperparameters as (b), but with Zellner's g-prior structure and

- (c) $g_0 = 0.1$
- (d) $g_0 = 1$
- (e) $g_0 = 10$

```
g_values <- c(c = 0.1, d = 1, e = 10)

# Typical g-prior: Lambda0 = g0^{-1} X'X  (confirm exact form in lecture)
# Then reuse the conjugate update from (b).

for (nm in names(g_values)) {
  g0 <- g_values[[nm]]
  # Lambda0_g <- (1/g0) * XtX
  # ... update and store results ...
  cat("Case", nm, ": g0 =", g0, "\n")
}
```

```
## Case c : g0 = 0.1
## Case d : g0 = 1
## Case e : g0 = 10
```

Answers (c): …

Answers (d): …

Answers (e): …

(f) Comment on differences

Write your discussion here. How does the prior influence the estimates across (a)–(e)?

Problem 4 (10 points): Prior / Likelihood / Posterior Summaries

For each setting, briefly describe (i) prior, (ii) likelihood, (iii) posterior, including interpretation of parameters.

(a) Beta–Binomial

(i) Prior: …
(ii) Likelihood: …
(iii) Posterior: …

(b) Poisson data

(i) Prior: …
(ii) Likelihood: …
(iii) Posterior: …

(c) Normal distribution

(i) Prior: …
(ii) Likelihood: …
(iii) Posterior: …

Problem 5 (20 points): Readings Discussion

Papers (on Canvas):

1. Rossi & Allenby (1993)
2. Rossi, McCulloch, & Allenby (1996)
3. Montgomery (1997)
4. Montgomery & Rossi (1999)
5. Rossi & Allenby (2003)

Note: Empirical Bayes Hierarchical Bayes (the course treats the latter as "real" Bayes). Cite papers in your answers.

(a) Random effects vs fixed effects vs Bayesian (e.g. price elasticities)

…

(b) Scanner-panel information for household heterogeneity; how to incorporate it

…

(c) Why Bayes is useful for household heterogeneity / what data limits it addresses

…

(d) How marketing researchers use / interpret the prior

…

(e) Hierarchical Bayes and MCMC (why MCMC is needed)

…

Session info

```r
sessionInfo()
```

```
## R version 4.3.1 (2023-06-16)
## Platform: aarch64-apple-darwin20 (64-bit)
## Running under: macOS 26.6.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
##
```

```
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## loaded via a namespace (and not attached):
##  [1] compiler_4.3.1    fastmap_1.2.0     cli_3.6.5         tools_4.3.1
##  [5] htmltools_0.5.8.1 yaml_2.3.10       tinytex_0.57      rmarkdown_2.29
##  [9] knitr_1.50        xfun_0.52         digest_0.6.37     rlang_1.1.6
## [13] evaluate_1.0.4
```